

MAE 4730
Lecture 33
11/12/2018

Today:

- Linear operators (cont'd)
- Rotation tensor (review)
- Rotation is linear
- Axis-angle

Linear Operators:

Superposition: "sum of the inputs is the output of the sums"

$$\mathcal{L}(av + bw) = a\mathcal{L}(v) + b\mathcal{L}(w)$$

scalar valued functions of a scalar: $\mathcal{L}(v) = cv$

scalar valued functions of a vector: $\mathcal{L}(\vec{v}) = c_i v_i = \underbrace{\vec{c} \cdot \vec{v}}_{\mathcal{L}(\vec{v})}$

Generalize:

$\vec{\mathcal{L}}(\vec{v})$ vector valued function of $\vec{v}$

Look at what $\vec{\mathcal{L}}(\vec{v})$ does to $\hat{e}_i$

$$\vec{\mathcal{L}}(\hat{e}_i) = \vec{L}_i$$

$$\vec{\mathcal{L}}(\vec{v}) = \vec{\mathcal{L}}(v_i \hat{e}_i) = v_i \vec{\mathcal{L}}(\hat{e}_i)$$

$$= \vec{L}_i v_i$$

$$\vec{L} \cdot \vec{v} = \hat{e}_i \cdot \vec{v}$$

$$\vec{\mathcal{L}}(\vec{v}) = \underbrace{\vec{L}_i \hat{e}_i \cdot \vec{v}}_{\vec{\mathcal{L}}(\vec{v})}$$

$\vec{L}_i \hat{e}_i = \text{dyadic representation of } \vec{\mathcal{L}}$

Rotations:

object in
reference coordinates



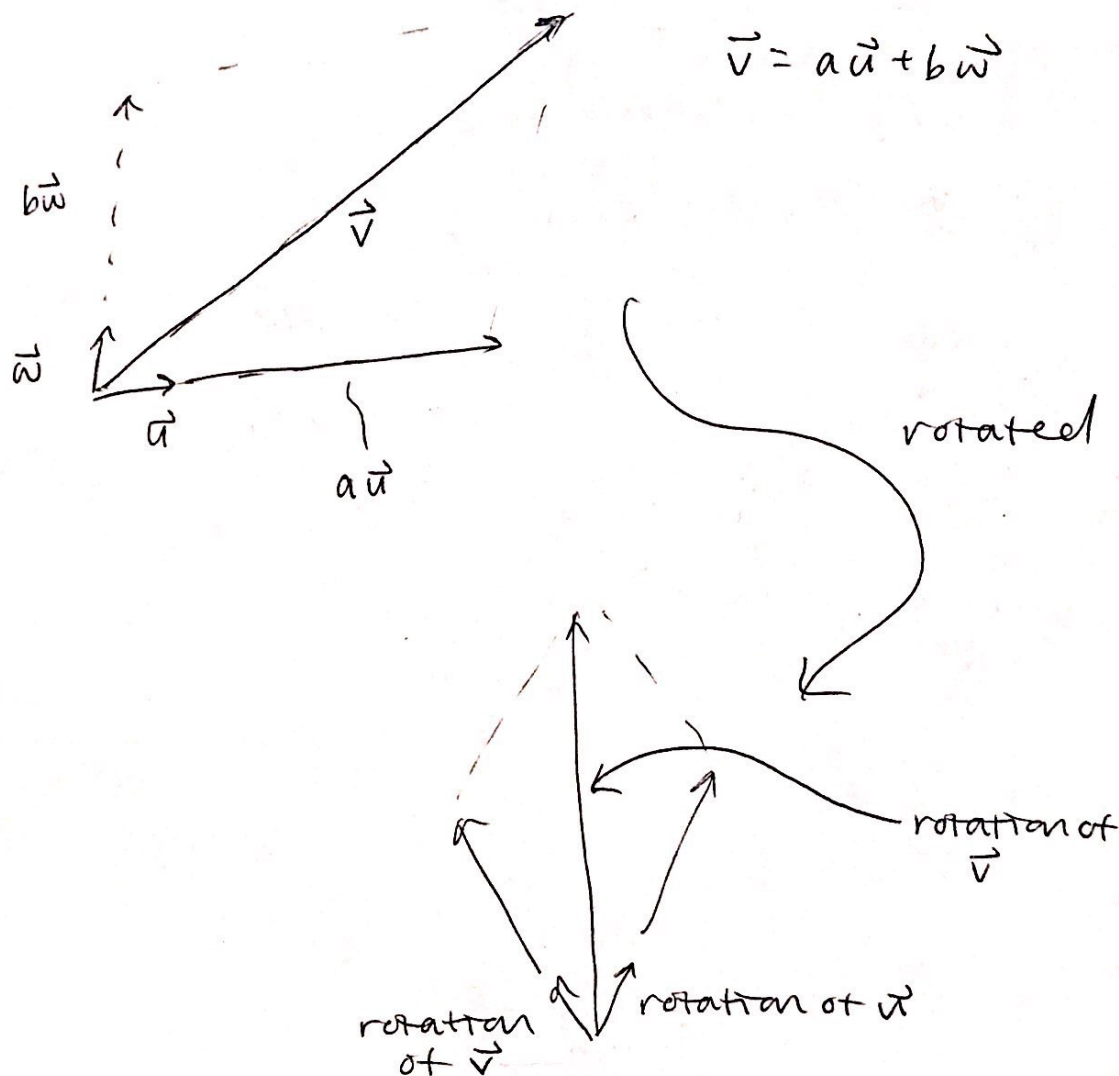
rotate



transformed

rotated collection
of vectors

collection of
vectors

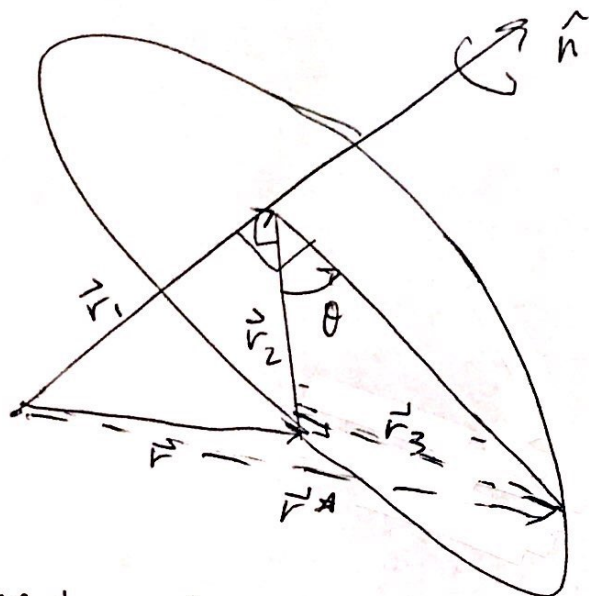


$$\Rightarrow \text{rotation of } (a\vec{u} + b\vec{w}) = a \text{Rot}(\vec{u}) + b \text{Rot}(\vec{w})$$

$\Rightarrow$ Rotation is a linear function

$$\Rightarrow \underline{\underline{R = \hat{e}_i' \hat{e}_i}}$$

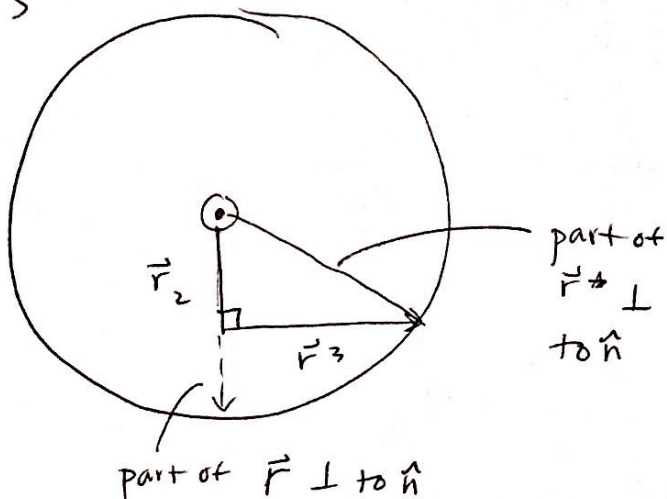
$$\hat{e}_i' \hat{e}_j' \xrightarrow{\text{rot of } \hat{e}_i = \hat{e}_i'} = R_{ij} \hat{e}_i \hat{e}_j$$



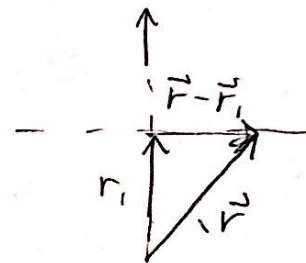
Given $\hat{n}, \theta, \vec{r}$, find $\vec{r}^*$

$$\vec{r}^* = \vec{r}_1 + \vec{r}_2 + \vec{r}_3$$

Looking down on
 $\hat{n}$ axis



Side view:



$$\vec{r}_1 = (\vec{r} \cdot \hat{n}) \hat{n} = \hat{n} (\hat{n} \cdot \vec{r})$$

$$\vec{r}_2 = (\vec{r} - \vec{r}_1) \cos \theta$$

$$\vec{r}_3 = (\hat{n} \times \vec{r}) \sin \theta$$

$$\vec{r}^* = \vec{r}_1 + \vec{r}_2 + \vec{r}_3$$

$$\boxed{\vec{r}^* = \hat{n} (\hat{n} \cdot \vec{r}) + (\vec{r} - \hat{n} (\hat{n} \cdot \vec{r})) \cos \theta + \hat{n} \times \vec{r} \sin \theta}$$

Rotation of $(\vec{r}) = \underline{\underline{R}} \cdot \vec{r}$

$$= (\hat{n}\hat{n}) \cdot \vec{r} + \cos\theta (\underline{\underline{I}} - \hat{n}\hat{n}) \vec{r} + \underbrace{\sin\theta \hat{n} \times \vec{r}}$$

$$\hookrightarrow \mathcal{L}(\hat{n}) \cdot \vec{r} = \hat{n} \times \vec{r}$$

AXIS
ANGLE
FORMULA

$$\boxed{\vec{r}^* = \underbrace{[\hat{n}\hat{n} + \cos\theta (\underline{\underline{I}} - \hat{n}\hat{n}) + \sin\theta \mathcal{L}(\hat{n})]}_{\underline{\underline{R}}} \cdot \vec{r}}$$

What is $\mathcal{L}(\hat{n})$

$\underbrace{\hspace{1cm}}$ skew symmetric tensor associated w/ $\hat{n}$

$$\mathcal{L}(\hat{n}) \cdot \vec{a} = \hat{n} \times \vec{a}$$

$$\left[\mathcal{L}(\hat{n}) \right]_F \cdot \left[\vec{r} \right]_F = \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$= n_2 r_3 - r_1$$